

REAL-TIME DETECTION METHOD FOR CRACKED EGGS BASED ON IMPROVED YOLOV8 AND DUAL VERIFICATION WITH TRIPLE CLASSIFICATION GRANULARITY

CSQ413 Seminar Presentation

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Outline

- 1 Introduction and Problem Statement
- 2 Literature Survey
- 3 System Hardware and Data Acquisition
- 4 Proposed Improved YOLOv8 Methodology
- 5 Experimental Setup and Evaluation
- 6 Results and Performance Analysis
- 7 Discussion and Conclusion

Introduction: Background and Motivation

- Egg shell integrity is a critical quality indicator in the automated poultry industry.
- Approximately 3% of eggs suffer shell damage during production, handling and transit.
- Damaged shells cause bacterial infiltration, accelerate spoilage, lead to yolk leakage and produce heavy economic losses.
- Commercial egg inspection in China still primarily relies on manual visual examination:
 - High labor intensity and high operating costs.
 - Low inspection efficiency and strong subjectivity.
 - Rapid worker fatigue causes elevated error rates and missed detections.

Limitations of Existing Methods

- **Acoustic Resonance Techniques:**

- Analyze resonance frequencies from mechanical tapping response.
- Highly sensitive to loud noise in factory environments. High hardware cost.

- **Traditional Computer Vision Approaches:**

- Depend on hand-crafted visual features (Otsu thresholding, morphological operations).
- Slow processing speed, weak adaptability and sensitivity to lighting variations.

- **Existing Deep Learning Models:**

- High network complexity and serial processing limit frame rates below production demands (≥ 30 FPS).
- Primarily evaluated on static, pre-washed egg images.
- Lack rotational conveyor mechanisms, leaving blind spots on the bottom of eggs.

Research Objectives and Key Contributions

Research Objective: Establish an industrial-grade, real-time online detection method for unwashed cracked eggs on dynamic conveyor lines.

Core Contributions of the Study:

- 1 **Triple Granularity and Dual Discrimination:** Formulated a fine-grained category scheme (intact egg, cracked egg, crack region) and a dual verification rule to prevent missed detections.
- 2 **Structural Optimization of YOLOv8:** Integrated DCNv2, BiFPN, WIoUv2 loss function and an extra small object detection head to capture minute, irregular cracks.
- 3 **Industrial Deployment and Validation:** Implemented end-to-end online inference on a Kyowa/FP-600 egg conveyor line, achieving 45 FPS and meeting industrial reliability standards.

• Traditional and Two-Stage Methods:

- **Liang et al. (2020):** Otsu threshold segmentation achieved 98.18% accuracy, but relied on manual feature extraction and static conditions.
- **Datta et al. (2019):** Faster R-CNN achieved 75% mAP for severe cracks; heavy two-stage computation was too slow for production lines.
- **Turkoglu (2021):** Hybrid CNN-BiLSTM achieved 96.3% accuracy, but incurred high computational complexity.
- **Botta et al. (2022):** Deep CNN reached 95.38% accuracy on static egg images.

• YOLO Adaptations and Real-Time Systems:

- **Yin et al. (2024):** YOLOv5 with Ghost and ECA modules for duck eggs (93.8% accuracy).
- **Huang et al. (2023):** YOLOv5 with BiFPN and GSConv on unwashed eggs (96.4% accuracy).

• Critical Research Gaps Addressed by Base Paper:

- Non-rotating conveyors fail to inspect bottom cracks.
- Binary classification (intact vs cracked) misses subtle micro-cracks under dynamic motion.

Machine Vision and Conveying System Setup

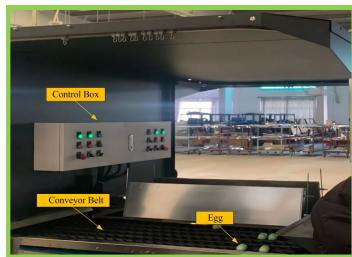


Figure: Egg conveying device (Source: Figure 1)

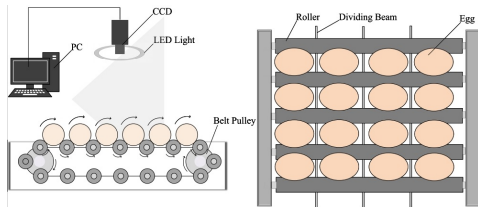


Figure: Machine vision system layout (Source: Figure 2)

- **Compound Motion:** Six parallel channels (360 eggs/min). Driven rollers rotate eggs 360 degrees, presenting bottom surfaces to the camera.
- **Imaging Rig:** CCD camera plane is parallel to egg motion plane; regulated LED lighting.

Dataset Acquisition and Preparation

- **On-Site Dynamic Image Acquisition:**

- Acquired at Guangzhou Suiping Enterprise Co., Ltd.
- Captured unwashed eggs with dust, stains, varying angles and real factory lighting.
- Deliberately collected cracked eggs to balance positive and negative samples.
- Base raw dataset: 628 dynamic images.

- **Dataset Splitting and Data Augmentation:**

- Split into Training (377), Validation (126) and Test (125) sets in a 6:2:2 ratio.
- Applied random cropping and horizontal/vertical flipping (50% probability) to training set.
- Expanded training set to 2,972 images after invalid data removal.
- Final dataset: 3,222 images containing 7,611 intact eggs, 2,711 cracked eggs and 2,935 cracked areas.

Triple Classification Granularity and Annotation

- **Limitation of Binary Classification:**

- Binary labels ("intact" vs "cracked") force the model to rely only on whole-egg features.
- Small local cracks contribute minor global visual cues, causing high false negatives.

- **Proposed Triple Classification Scheme:**

- **Category 1 (Intact Egg):** Normal undamaged egg body.
- **Category 2 (Cracked Egg):** Entire egg body displaying cracks.
- **Category 3 (Crack Region):** Specific localized damaged fracture zone.

- **Annotation Protocol:**

- Annotated using Labellmg with labels "1", "2" and "3".
- XML annotations converted to YOLO-compatible TXT format.
- Enriches feature representations with both macro-level and micro-level cues.

Improved YOLOv8 Network Architecture Overview

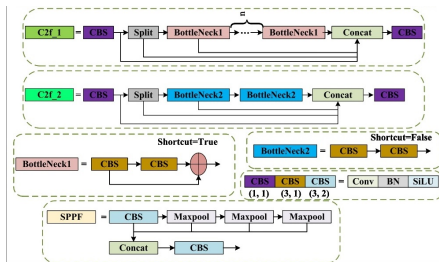
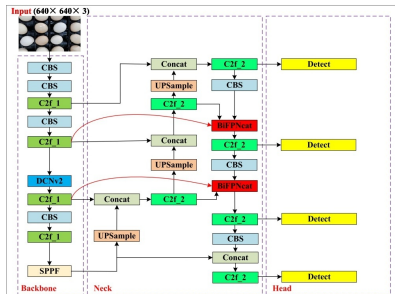


Figure: Architecture of the proposed improved YOLOv8 model (Source: Base Paper Figure 3)

- Retains anchor-free split Ultralytics head for optimal accuracy-speed trade-off.
- Integrates four dedicated innovations: DCNv2, BiFPN, Small Object Head and WIoUv2.

Architectural Enhancement 1: Deformable Convolutions (DCNv2)

Limitation of Standard Convolutions:

- Rigid square sampling grids cannot adapt to irregular, jagged crack contours.

DCNv2 Formulation:

$$Y(q) = \sum_{k=1}^K w_k X(q + q_k + \Delta q_k) \Delta m_k$$

- Δq_k : Learnable 2D spatial offset matching crack geometry.
- $\Delta m_k \in [0, 1]$: Dynamic modulation scalar scaling feature amplitude.
- Amplifies crack features while suppressing background shell noise.

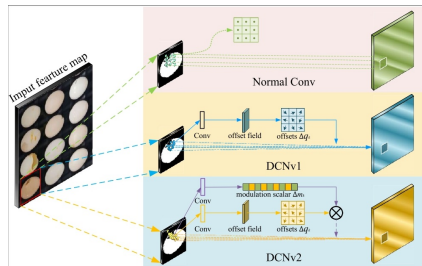


Figure: Sampling comparison: Normal Conv, DCNv1, DCNv2 (Source: Figure 4)

Architectural Enhancement 2: BiFPN Multi-Scale Fusion

From FPN and PANet to BiFPN:

- **FPN:** Top-down path for semantic feature integration.
- **PANet (Default YOLOv8):** Added bottom-up path for low-level spatial features.
- **BiFPN:** Removes single-edge nodes, adds cross-scale skip connections and repeats blocks.

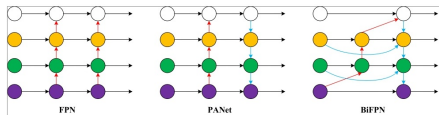


Figure: Architectural comparison of FPN, PANet and BiFPN (Source: Base Paper Figure 5)

Key Benefits for Crack Inspection:

- Dynamically weights feature maps based on contribution.
- Maintains strong representation under lighting variations and dirty shell backgrounds.

Enhancements 3 & 4: Small Target Head and WIoUv2 Loss

• Enhancement 3: Dedicated Small Object Detection Layer

- Standard YOLOv8 5 downsamplings ($/32$) yield an 80×80 head with receptive field 8×8 px; micro-cracks under 8 px vanish in deep layers.
- Integrated a 4th detection head operating on a 160×160 feature map (P2 layer).
- Directly captures fine spatial textures and edge gradients from earlier stages.

• Enhancement 4: Wise-IoU v2 (WIoUv2) Loss Function

$$L_{\text{WIoUv2}} = L_{\text{IoU}}^{\gamma*} L_{\text{WIoUv1}}, \quad \gamma > 0$$

- Standard CloU ignores sample quality variations during dynamic conveyor motion.
- Dynamic focusing factor $L_{\text{IoU}}^{\gamma*}$ adaptively balances gradient contributions.
- Handles class imbalance and enhances detection of occluded or motion-blurred cracks.

Dual Verification Decision-Making Mechanism

- **Post-Processing Decision Rule:**

- Standard sorting systems decide based only on whole-egg bounding boxes.
- Under the proposed rule, an egg is tagged as **damaged** if the detector outputs:

Category 2 (Cracked Egg) OR Category 3 (Crack Region)

- **Mathematical Joint Miss Rate Reduction:**

- A damaged egg is missed **only if both** Category 2 and Category 3 fail simultaneously:

$$P(\text{Miss}_{\text{Overall}}) = P(\text{Miss}_{\text{Egg}}) \times P(\text{Miss}_{\text{Region}})$$

- Global shape cues and local fracture cues cross-verify each other.
- Lowers industrial false negative rates while preventing unnecessary waste.

Experimental Platform and Training Setup

Hardware and Software

Specifications:

- **Operating System:** Windows 10
- **Framework:** Python 3.8, PyTorch 1.13.1, CUDA 11.6
- **Processor:** Intel Core i5-10400F @ 2.9 GHz
- **Graphics Card:** NVIDIA GeForce RTX 3060 (12 GB VRAM)
- **System Memory:** 32 GB DDR4 (3200 MHz)

Hyperparameters:

- Input resolution: 640×640 pixels
- Epochs: 300 iterations; Batch size: 16

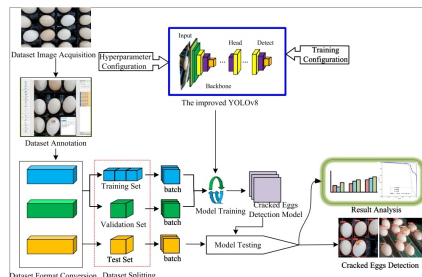


Figure: Training and validation workflow (Source: Base Paper Figure 6)

Evaluation Metrics

- **Precision (P):** Proportion of detected cracked items that are truly cracked:

$$\text{Precision} = \frac{TP}{TP + FP}$$

- **Recall (R):** Proportion of actual cracked items correctly detected:

$$\text{Recall} = \frac{TP}{TP + FN}$$

- **Mean Average Precision (mAP@0.5):** Comprehensive accuracy score computed as the area under the PR curve across all categories.
- **Frames Per Second (FPS):** Real-time processing speed (≥ 30 FPS needed on lines).
- **Industrial Reliability Metrics:**
 - **False Negative Rate (FNR):** Ratio of missed damaged eggs (product safety).
 - **False Positive Rate (FPR):** Ratio of good eggs falsely rejected (waste control).

Ablation Study and Model Complexity

Table: Ablation experiment results on test dataset (Source: Base Paper Table 1)

DCNv2	BiFPN	WIoU	Small Target	Precision (%)	Recall (%)	mAP:0.5 (%)
-	-	-	-	92.2	85.3	87.7
-	-	✓	-	90.6	87.4	90.3
-	✓	-	-	89.8	83.8	88.2
-	-	-	✓	93.8	90.3	93.0
✓	-	✓	-	93.1	89.4	92.1
-	✓	✓	-	92.6	89.5	92.2
✓	✓	✓	-	93.8	89.9	92.6
✓	✓	✓	✓	94.1	90.0	92.9

- Small target head yields the largest single mAP gain (+5.3%). Full ensemble achieves the highest overall Precision (94.1%) and balanced robustness.
- **Model Complexity (Table 2):** Parameters: 2.66×10^7 (+2.7%), FLOPs: 86.5 G (+9.3%), Model Size: 51.1 MB (+3.0%). Significant accuracy boost achieved with minimal overhead.

Comparative Experiments with Mainstream Detectors

Table: Comparison across different object detection models (Source: Base Paper Table 3)

Model	Precision (%)	Recall (%)	mAP:0.5 (%)	FPS
Faster R-CNN (ResNet50)	68.4	80.7	78.1	17
YOLOv5	86.9	81.1	85.2	58
YOLOv8 (Baseline)	92.2	85.3	87.7	52
YOLOv11	92.7	84.3	87.3	47
Our Proposed Model	94.1	90.0	92.9	45

- Outperformed Faster R-CNN by 14.8% mAP despite Faster R-CNN using pretrained weights while our model trained from scratch.
- Outperformed recent YOLOv11 by 5.6% mAP and baseline YOLOv8 by 5.2% mAP.
- At 45 FPS, detection speed easily satisfies the production requirement (≥ 30 FPS).

Laboratory Simulated Platform Testing

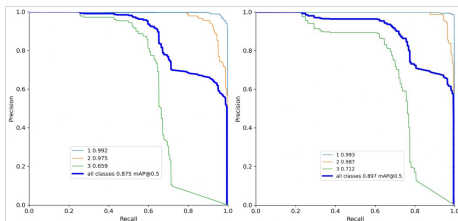


Figure: PR curves comparison (Source: Figure 8)



Figure: Visual detection comparison (Source: Figure 9)

- Tested on 115 video frames: improved model achieves 2.5% higher mAP.
- Successfully identifies micro-cracks where original YOLOv8 produced missed detections.

Industrial Deployment on Factory Equipment

On-Site Equipment (Kyowa/FP-600 Line):

- Conveyor velocity: 0.1 m/s.
- Camera: A3200CU000 (1920 × 1080) at 120 FPS.
- Host CPU: 12th Gen Intel i5-12400F (2.50 GHz).
- Host GPU: GeForce RTX 3060 Ti 8G; 16 GB RAM.
- Environment: Python 3.8, PyTorch 1.13.1, CUDA 11.6.

Operational Speed:

- Original YOLOv8: 52 FPS.
- Proposed Model: 45 FPS (fully real-time).

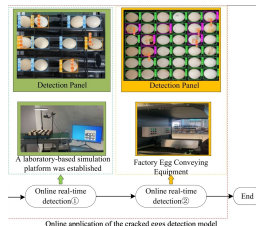
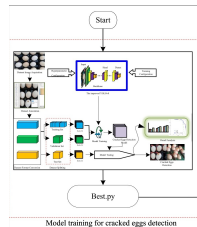


Figure: Online deployment on factory equipment (Source: Figure 7)

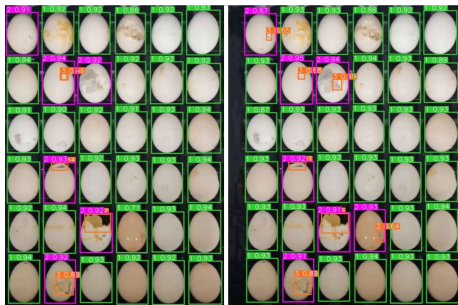
Factory Online Inspection Accuracy (FNR and FPR)

Table: Online inspection comparison on 106 production frames (Source: Base Paper Tables 5 and 6)

Category	Baseline YOLOv8		Proposed Improved YOLOv8	
	FNR (%)	FPR (%)	FNR (%)	FPR (%)
Class 1: Intact Egg	1.26	4.00	0.46	2.18
Class 2: Cracked Egg	10.17	3.20	5.52	1.16
Class 3: Damaged Area	22.97	0.87	19.19	0.58
Compound Dual Verification FNR			$10.17\% \times 22.97\% = 2.34\%$	
			$5.52\% \times 19.19\% = 1.06\%$	

- False alarm rate for intact eggs dropped from 4.00% to 2.18%, saving intact eggs.
- Compound miss rate for cracked eggs fell to 1.06%, reliably filtering defective eggs.

Production Line Visual Detections and Speed Stability



(a) YOLOv8 model detection results

(b) Improved YOLOv8 detection results

Figure: Detection on actual production line (Source: Figure 10)

- Stable inference performance across continuous 400-frame video testing.
- Accurately detects cracked eggs and damaged areas under variable industrial lighting.

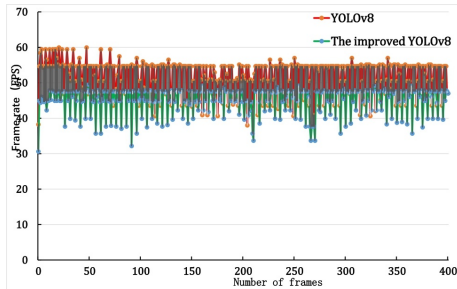


Figure: Speed stability across 400 frames (Source: Figure 11)

Discussion: Operational Findings and Edge Cases

- **Speed vs Accuracy Trade-off:**

- Inference speed decreased from 52 FPS to 45 FPS (-14.5%) due to added layers.
- 45 FPS remains well above the 30 FPS factory threshold, making the 5.2% mAP boost highly advantageous.

- **Adaptability to Eggshell Surface Conditions:**

- Successfully validated on both unwashed production eggs and pre-washed supermarket eggs.

- **Challenging Edge Cases Identified in the Study:**

- **Solid shells (e.g., duck eggs):** High accuracy due to distinct color and gloss changes.
- **Patterned shells (e.g., quail eggs):** Intricate natural spots create visual interference.
- **Fine net-like cracks:** Minute pixel footprint and faint gradients make them difficult to distinguish from shell textures.

Conclusion and Future Scope

Conclusion:

- Established an end-to-end real-time cracked egg inspection system combining rotational conveyance, improved YOLOv8 and dual verification.
- Integrated DCNv2, BiFPN, WIoUv2 and a P2 small target head, achieving 92.9% mAP (+5.2% over baseline) and surpassing Faster R-CNN, YOLOv5 and YOLOv11.
- Attained 45 FPS and lowered compound broken egg miss rate to 1.06% on factory equipment, satisfying industrial online sorting criteria.

Future Scope:

- Adopt dark-box illumination to increase visibility of low-contrast net-like cracks.
- Expand datasets to other avian eggs and integrate internal defect detection (scattered yolk, blood spots).

References



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Thank You!

Questions and Discussion